

Chapter 10

Basics of Neural Network

OBJECTIVE OF THE CHAPTER :

In the last 9 chapters, you have been introduced to the concepts of machine learning in great details. You started with how to decide whether a problem can be solved with machine learning, and once you have made that decision, you learnt how to start with the modelling of the problem in the machine learning paradigm. In that context, you were introduced to three types of machine learning – supervised, unsupervised, and reinforcement. Then, you explored all different popular algorithms of supervised and unsupervised learning. Now, you have gained quite some background on machine learning basics, and it is time for you to get introduced to the concept of neural network.

Well, you have already seen right at the beginning of the book how the machine learning process maps with the human learning process. Now, it is time to see how the human nervous system has been mimicked in the computer world in the form of an artificial neural network or simply a neural network. This chapter gives a brief view of neural networks and how it helps in different forms of learning.

10.1 INTRODUCTION

In the previous chapters, we were slowly unveiling the mystery of machine learning, i.e. how a machine learns to perform tasks and improves with time from experience, by expert guidance or by itself. Machine learning, as we have seen, mimics the human form of learning. On the other hand, human learning, or for that matter every action of a human being, is controlled by the nervous system. In any human being, the nervous system coordinates the different actions by transmitting signals to and from different parts of the body. The nervous system is constituted of a special type of cell, called **neuron** or **nerve cell**, which has special structures allowing it to receive or send signals to other neurons. Neurons connect with each other to transmit signals to or receive signals from other neurons. This structure essentially forms a network of neurons or a neural network.

By virtue of billions of networked neurons that it possesses, the biological neural network is a massively large and complex parallel computing network. It is because of this massive parallel computing network that the nervous system helps human beings to perform actions or take decisions at a speed and with such ease that the fastest supercomputer of the world will also be envy of. For example, let us think of the superb flying catches taken by the fielders in the cricket world cup. It is a combination of superior calculation based on past cricketing experience, understanding of local on-ground conditions, and anticipation of how hard the ball has been hit that the fielder takes the decision about when to jump, where to jump, and how much to jump. This is a highly complex task, and you may think that not every human being is skilled enough for such a magical action. In that case, let us think of

something much simpler. Let us consider about a very simple, daily activity, namely swimming in the pool. Apparently, swimming may look very trivial, but think about the parallel actions and decisions that need to be taken to swim. It needs the right combination of body position, limb movement, breathe in/out, etc. to swim. To add to the challenge is the fact that water is few hundred times denser than air. So, to orient the body movement in that scale difference is a difficult proposition in itself. Coordinating the actions and taking the decisions for such a complex task, which may appear to be trivial, are possible because of the massive parallel complex network, i.e. the neural network.

The fascinating capability of the biological neural network has inspired the inception of artificial neural network (ANN). An ANN is made up of artificial neurons. In its most generic form, an ANN is a machine designed to model the functioning of the nervous system or, more specifically, the neurons. The only difference is that the biological form of neuron is replicated in the electronic or digital form of neuron. Digital neurons or artificial neurons form the smallest processing units of the ANNs. As we move on, let us first do a deep dive into the structure of the biological neuron and then try to see how that has been modelled in the artificial neuron.

10.2 UNDERSTANDING THE BIOLOGICAL NEURON

The human nervous system has two main parts –

- the central nervous system (CNS) consisting of the brain and spinal cord
- the peripheral nervous system consisting of nerves and ganglia outside the brain and spinal cord.

The CNS integrates all information, in the form of signals, from the different parts of the body. The peripheral nervous system, on the other hand, connects the CNS with the limbs

and organs. Neurons are basic structural units of the CNS. A neuron is able to receive, process, and transmit information in the form of chemical and electrical signals. Figure 10.1 presents the structure of a neuron. It has three main parts to carry out its primary functionality of receiving and transmitting information:

1. **Dendrites** – to receive signals from neighbouring neurons.
2. **Soma** – main body of the neuron which accumulates the signals coming from the different dendrites. It ‘fires’ when a sufficient amount of signal is accumulated.
3. **Axon** – last part of the neuron which receives signal from soma, once the neuron ‘fires’, and passes it on to the neighbouring neurons through the axon terminals (to the adjacent dendrite of the neighbouring neurons).

There is a very small gap between the axon terminal of one neuron and the adjacent dendrite of the neighbouring neuron. This small gap is known as **synapse**. The signals transmitted through synapse may be excitatory or inhibitory.

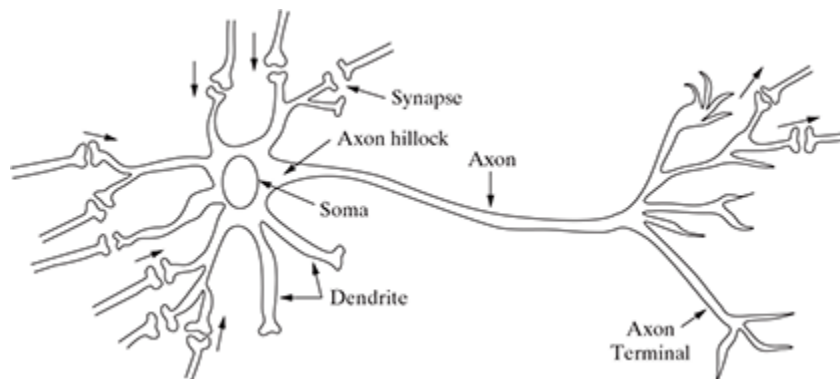


FIG. 10.1 Structure of biological neuron

Points to Ponder:

The adult human brain, which forms the main part of the central nervous system, is approximately 1.3 kg in weight and 1200 cm³ in volume. It is estimated to contain about 100 billion (i.e. 10¹¹) neurons and 10 times more glial or glue cells. Glial cells act as support cells for the neurons. It is believed that neurons represent about 10% of all cells in the brain. On an average, each neuron is connected to 10⁵ of other neurons, which means that altogether there are 10¹⁶ connections.

The axon, a human neuron, is 10–12 µm in diameter. Each synapse spans a gap of about a millionth of an inch wide.

10.3 EXPLORING THE ARTIFICIAL NEURON

The biological neural network has been modelled in the form of ANN with artificial neurons simulating the function of biological neurons. As depicted in [Figure 10.2](#), input signal x_i ($x_1, x_2, \dots, x_n$) comes to an artificial neuron. Each neuron has three major components:

1. A set of '*i*' **synapses** having weight w_i . A signal x_i forms the input to the *i*-th synapse having weight w_i . The value of weight w_i may be positive or negative. A positive weight has an excitatory effect, while a negative weight has an inhibitory effect on the output of the summation junction, y_{sum} .
2. A **summation junction** for the input signals is weighted by the respective synaptic weight. Because it is a linear combiner or adder of the weighted input signals, the output of the summation junction, y_{sum} , can be expressed as follows:

$$y_{\text{sum}} = \sum_{i=1}^n w_i x_i$$

[Note: Typically, a neural network also includes a bias which adjusts the input of the activation function. However, for the sake of simplicity, we are ignoring bias for the time being. In the case of a bias 'b', the value of y_{sum} would have been as follows:

$$y_{\text{sum}} = b + \sum_{i=1}^n w_i x_i$$

3. A **threshold activation function** (or simply **activation function**, also called **squashing function**) results in an output signal only when an input signal exceeding a specific threshold value comes as an input. It is similar in behaviour to the biological neuron which transmits the signal only when the total input signal meets the firing threshold.

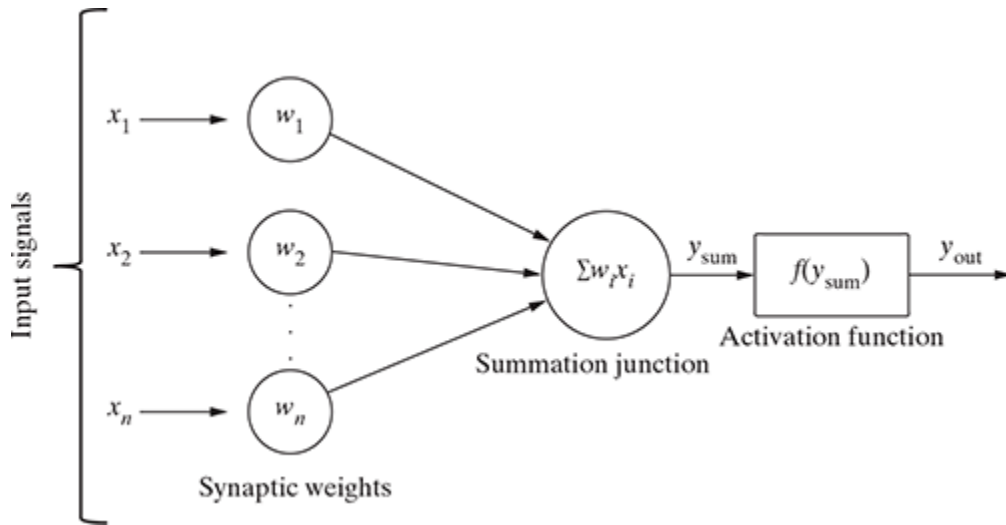


FIG. 10.2 Structure of an artificial neuron

Output of the activation function, y_{out} , can be expressed as follows:

$$y_{\text{out}} = f(y_{\text{sum}})$$

10.4 TYPES OF ACTIVATION FUNCTIONS

There are different types of activation functions. The most commonly used activation functions are highlighted below.

10.4.1 Identity function

Identity function is used as an activation function for the input layer. It is a linear function having the form

$$y_{\text{out}} = f(x) = x, \text{ for all } x$$

As obvious, the output remains the same as the input.

10.4.2 Threshold/step function

Step/threshold function is a commonly used activation function. As depicted in Figure 10.3a, **step function** gives 1 as output if the input is either 0 or positive. If the input is negative, the step function gives 0 as output. Expressing mathematically,

$$y_{\text{out}} = f(y_{\text{sum}}) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

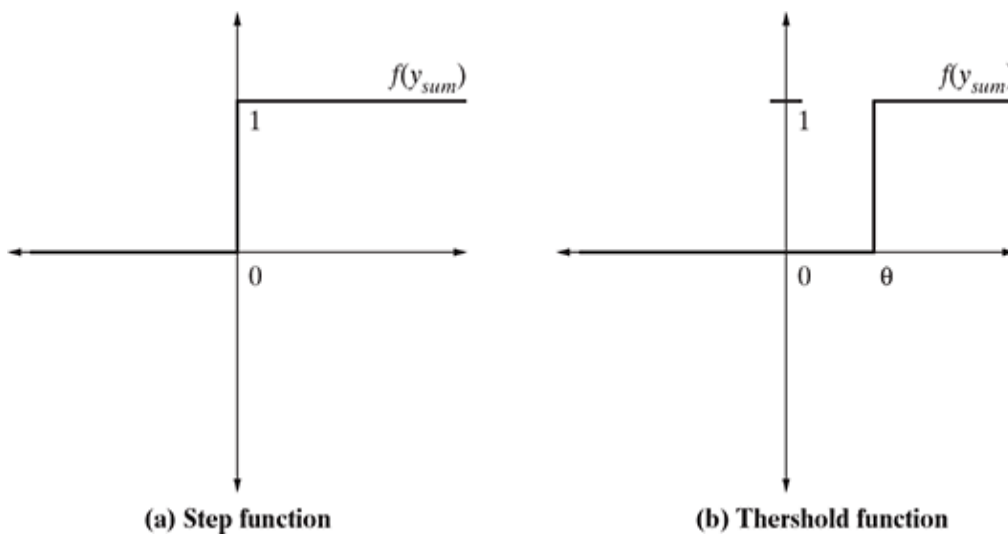


FIG. 10.3 Step and threshold functions

The **threshold function** (depicted in [Fig. 10.3b](#)) is almost like the step function, with the only difference being the fact that θ is used as a threshold value instead of 0. Expressing mathematically,

$$y_{\text{out}} = f(y_{\text{sum}}) = \begin{cases} 1, & x \geq \theta \\ 0, & x < \theta \end{cases}$$

10.4.3 ReLU (Rectified Linear Unit) function

ReLU is the most popularly used activation function in the areas of convolutional neural networks and deep learning. It is of the form

$$f(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

This means that $f(x)$ is zero when x is less than zero and $f(x)$ is equal to x when x is above or equal to zero. [Figure 10.4](#) depicts the curve for a ReLU activation function.

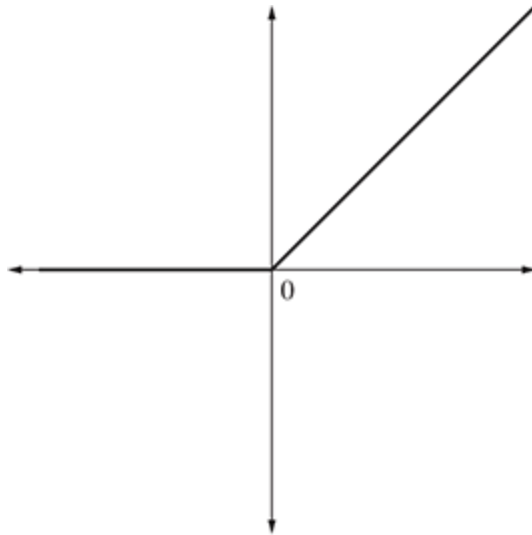


FIG. 10.4 ReLU function

This function is differentiable, except at a single point $x = 0$. In that sense, the derivative of a ReLU is actually a sub-derivative.

10.4.4 Sigmoid function

Sigmoid function, depicted in [Figure 10.5](#), is by far the most commonly used activation function in neural networks. The need for sigmoid function stems from the fact that many learning algorithms require the activation function to be differentiable and hence continuous. Step function is not suitable in those situations as it is not continuous. There are two types of sigmoid function:

1. Binary sigmoid function
2. Bipolar sigmoid function

10.4.4.1 Binary sigmoid function

A binary sigmoid function, depicted in [Figure 10.5a](#), is of the form

$$y_{\text{out}} = f(x) = \frac{1}{1 + e^{-kx}}$$

where k = steepness or slope parameter of the sigmoid function. By varying the value of k , sigmoid functions with different slopes can be obtained. It has range of (0, 1).

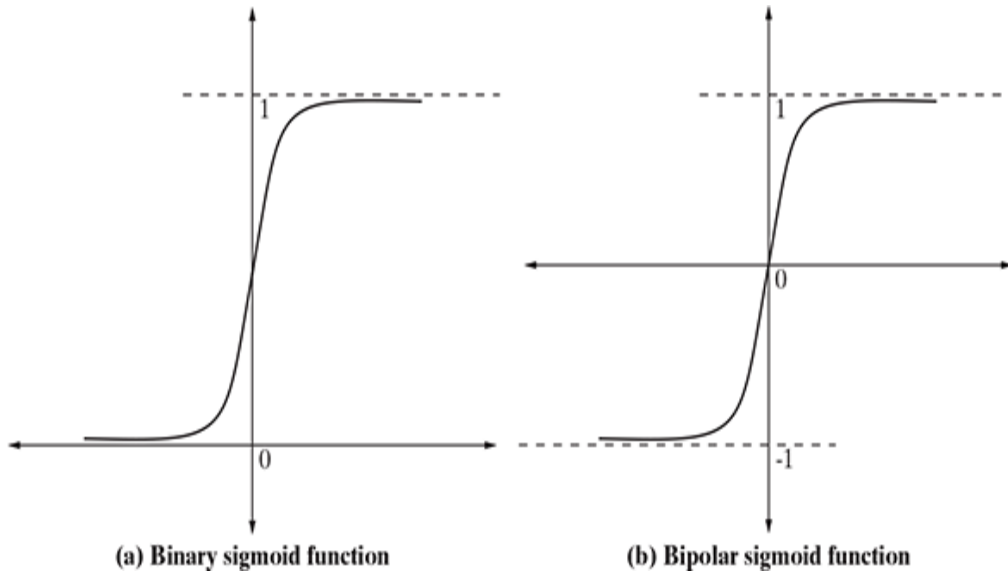


FIG. 10.5 Sigmoid function

The slope at origin is $k/4$. As the value of k becomes very large, the sigmoid function becomes a threshold function.

10.4.4.2 Bipolar sigmoid function

A bipolar sigmoid function, depicted in Figure 10.5b, is of the form

$$y_{\text{out}} = f(x) = \frac{1 - e^{-kx}}{1 + e^{-kx}}$$

The range of values of sigmoid functions can be varied depending on the application. However, the range of $(-1, +1)$ is most commonly adopted.

10.4.5 Hyperbolic tangent function

Hyperbolic tangent function is another continuous activation function, which is bipolar in nature. It is a widely adopted activation function for a special type of neural network known as backpropagation network (discussed elaborately in Section 10.8). The hyperbolic tangent function is of the form

$$y_{\text{out}} = f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

This function is similar to the bipolar sigmoid function.

Note that all the activation functions defined in Sections 10.4.2 and 10.4.4 have values ranging between 0 and 1. However, in some cases, it is desirable to have values ranging from -1 to $+1$. In that case, there will be a need to reframe the activation function. For example, in the case of step function, the revised definition would be as follows:

$$y_{\text{out}} = f(y_{\text{sum}}) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

Did you know?

Works related to neural network dates long back to the 1940s. Warren McCulloch and Walter Pitts in 1943 created a computational network inspired by the biological processes in the brain. This model paved the way for neural networks.

The next notable work in the area of neural network was in the late 1940s by the Canadian psychologist Donald Olding Hebb. He proposed a learning hypothesis based on the neurons and synaptic connection between neurons. This became popular as Hebbian learning.

In 1958, the American psychologist Frank Rosenblatt refined the Hebbian concept and evolved the concept of perceptron. It was almost at the same time, in 1960, that Professor Bernard Widrow of Stanford University came up with an early single-layer ANN named ADALINE (Adaptive Linear Neuron or later Adaptive Linear Element). Marvin Lee Minsky and Seymour Aubrey Papert stated two key issues of perceptron in their research paper in 1969. The first issue stated was the inability of perceptron of processing the XOR (exclusive-OR) circuit. The other issue was lack of processing power of computers to effectively handle the large neural networks.

From here till the middle of the 1970s, there was a slowdown in the research work related to neural networks. There was a renewed interest generated in neural networks and learning with Werbos' backpropagation algorithm in 1975. Slowly, as the computing ability of the computers increased drastically, a lot of research on neural network has been conducted in the later decades. Neural network